



SS-31 (Elamipretide)

Information sheet written for the 10 mg vial TrueHold holds in stock. Not a generic internet protocol.

THIS VIAL — STARTING DOSE (read this first)

SS-31 (Elamipretide) · 10 mg lyophilized · mix with 2 mL bacteriostatic water · 5 mg/mL · 1 unit = 50 mcg

Start: draw **20 units** on a U-100 syringe (**0.20 mL = 1 mg**). On a 0.5 mL (50-unit) U-100 syringe, draw to the 20 mark.

Route: Subcutaneous (belly or upper arm)

Rhythm: Daily for 5 days, then 2 days off, for 4 weeks, then a break. Not the Barth-syndrome pharmaceutical dose.

Starting draw: 20 units (1 mg = 0.20 mL)



1. What it is — the molecule, how large it is

SS-31 is a tiny mitochondria-targeted tetrapeptide (D-Arg-Dmt-Lys-Phe-NH₂), also called elamipretide. The hydrochloride salt used in the approved drug has molecular weight 749.2. TrueHold stocks a 10 mg lyophilized research vial — one-quarter of a single 40 mg Forzinity pharmaceutical dose.

Mitochondria — where NAD⁺, MOTS-c, and SS-31 work

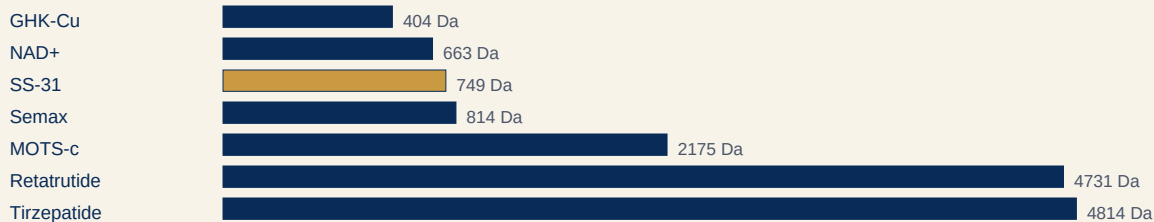


Inner membrane (cristae + cardiolipin) — SS-31 binds here

NAD⁺ shuttles electrons for ATP

MOTS-c is a 16-aa mitochondrial message to muscle / AMPK

How large is it? (daltons, log-ish bars — bigger bar = bigger molecule)



2. How it works in the body

It concentrates on the inner mitochondrial membrane and binds cardiolipin, the lipid that folds the cristae and holds the electron-transport chain together. The hope is better ATP production and less oxidative leak. It is not an incretin and not NAD⁺.



3. Testing data — animal

Many mitochondrial-disease and ischemia models show improved bioenergetics when SS-31 reaches mitochondria. That is a strong preclinical story and why Stealth BioTherapeutics took elamipretide into rare-disease trials.

4. Testing data — human, what is expected, hopes

On 19 September 2025 FDA granted accelerated approval to Forzinity (elamipretide) 40 mg subcutaneous daily to improve muscle strength in Barth syndrome (patients ≥ 30 kg), based on knee-extensor strength, with a confirmatory trial required. Common trial adverse events were injection-site reactions. TrueHold's 10 mg vial is not Forzinity and cannot deliver 40 mg/day.

- Cellular energy and stamina notes
- Mitochondrial-efficiency research
- A daily cardiolipin-targeted shot at a research-vial milligram, not a Barth-syndrome prescription

5. Have people hurt themselves? Risks

Forzinity labeling: injection-site reactions are common; serious reactions have been reported. The branded syringe also contains benzyl alcohol (a neonate toxicity issue — not for babies). Using a 10 mg research vial is not a Barth-syndrome treatment. Do not chase 40 mg/day by buying four vials and guessing. Stop for severe injection-site necrosis, breathing trouble, or dark urine.

6. FDA — approved or not

The molecule elamipretide IS FDA-approved as Forzinity for Barth syndrome only (accelerated approval, 2025). TrueHold's 10 mg lyophilized SS-31 vial is NOT Forzinity, NOT approved, and NOT dosed at 40 mg. This sheet's 1–2 mg start is vial-honest research handling, not the Barth prescription.

7. Reconstitution for this exact vial

Check the label: it must say SS-31 (Elamipretide) and 10 mg powder in a standard ~3 mL glass vial. You need bacteriostatic water (BAC), alcohol swabs, and U-100 insulin syringes (0.5 mL / 50-unit for every starting draw on this sheet; 1 mL / 100-unit if a later step exceeds 50 units).

- Wash hands. Wipe the rubber top with an alcohol swab.
- Draw **2 mL** of BAC into a mixing syringe.
- Insert through the stopper. Inject slowly down the inside glass wall — not straight onto the powder.
- Gently roll or swirl until clear. Do not shake.
- Wipe the top again. Write today's date on the vial. Refrigerate 36–46°F. Use within 28 days.
- Mix and store in this vial only. Do not transfer.

8. Loading the 0.5 mL syringe — schedule for this vial

U-100 math is the same on a 0.5 mL or 1 mL syringe: 100 units = 1.00 mL, so 50 units = 0.50 mL (a completely full 0.5 mL syringe). Starting draws on this sheet all fit 0.5 mL. The green row is the start.

When	Units (U-100)	mg	mL	0.5 mL syringe	How often
Weeks 1–2 (start here)	20	1	0.20	Draw to 20	Once daily, 5 days on / 2 days off
Weeks 3–4 (if tolerated)	40	2	0.40	Draw to 40	Once daily, 5 days on / 2 days off

- 10 mg in 2 mL $\rightarrow$ 5 mg/mL. 20 units = 1.0 mg; 40 units = 2.0 mg. Both fit a 0.5 mL syringe.
- FDA-approved Forzinity (elamipretide) is 40 mg subcutaneous daily for Barth syndrome in patients ≥ 30 kg. That is a different product. This 10 mg research vial cannot supply a 40 mg daily pharmaceutical dose.
- At 20 units five days per week one mixed vial lasts 2 weeks.

Fulfillment

- Prep and local delivery: Las Vegas residents only.
- Shipping: dry (lyophilized) vials only — we do not ship mixed liquid.
- Interest orders are confirmed by phone before payment.



Working with the team

- Share a phone number so the team can call.
- Confirm Las Vegas residency for prep and local delivery.
- Complete required documentation with the team.
- If anything on this sheet disagrees with your vial label or COA, stop and call before mixing.

Sources

- FDA Forzinity accelerated approval (19 Sep 2025):
<https://www.fda.gov/news-events/press-announcements/fda-grants-accelerated-approval-first-treatment-barth-syndrome>
- DailyMed Forzinity: 40 mg SC daily; MW 749.2 (HCl salt)
- TrueHold 10 mg / 2 mL: 20 units = 1 mg, 40 units = 2 mg
- TrueHold 10 mg vial / 2 mL BAC. Conservative research-vial start (1–2 mg), explicitly not Forzinity 40 mg.

Protocol note

TrueHold writes these sheets for the milligram vials we hold. Mix volume and unit marks were double-checked so milligrams, milliliters, and U-100 units agree. This is educational research information, not a prescription and not medical advice. The Telegram bot does not dose in chat — read this PDF. For an individualized plan, use /schedule. Research use only.

